

Introduction to Professional Practices in IT

CS4001 – Professional Practices in IT

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Investment Appraisal

CLASS ACTIVITY 7

<https://tinyurl.com/Fall2025-Activity7>

Activity 7

A small manufacturing company is deciding whether to **buy** or **rent** a new **industrial 3D printer**.

- **Purchase price:** £15,000 (paid immediately).
- **Annual insurance & registration:** £400 + £200 = **£600** per year.
- **Maintenance cost** is estimated as:
 - Year 0 (now): £300
 - Year 1: £300
 - Year 2: £400
 - Year 3: £500
 - Year 4: £600

At the end of Year 4 (i.e., after 5 years including Year 0), the printer is expected to be sold for **£3,000** (this resale is a future cash receipt already given — do not inflate it).

The company's borrowing/discount rate is **9%**.

Activity 7 Cont.

If the company **rents** a printer instead: cost = **£60 per day**, and they need it **60 days per year** → **£3,600 per year**.

Inflation is 4% per year (all annual costs inflate at 4% per year; resale value is the actual future cash amount and is not inflated further).

Time horizon: Years 0, 1, 2, 3, 4 (Year 0 is now).

Decide which option (buy or rent) is cheaper in present value terms using DCF.

STEP 1: Inflate the repeating annual costs (4% p.a.)

We compute the nominal cost that will be paid in each year by applying inflation to the base amounts.

Insurance + registration (base £600):

Year 0: £600

Year 1: $600 \times 1.04 = \textbf{£624}$

Year 2: $600 \times 1.04^2 = \textbf{£649}$

Year 3: $600 \times 1.04^3 = \textbf{£675}$

Year 4: $600 \times 1.04^4 = \textbf{£702}$

STEP 1: Cont.

Maintenance (use the year-specific base then inflate from now to that year):

Year 0: £300

Year 1: $300 \times 1.04 = \mathbf{£312}$

Year 2: $400 \times 1.04^2 = 400 \times 1.0816 = \mathbf{£433}$

Year 3: $500 \times 1.04^3 = 500 \times 1.124864 = \mathbf{£562}$

Year 4: $600 \times 1.04^4 = 600 \times 1.169859 = \mathbf{£702}$

STEP 1: Cont.

Rent cost (base £3,600):

Year 0: £3,600

Year 1: $3,600 \times 1.04 = \textbf{£3,744}$

Year 2: $3,600 \times 1.04^2 = \textbf{£3,894}$

Year 3: $3,600 \times 1.04^3 = \textbf{£4,050}$

Year 4: $3,600 \times 1.04^4 = \textbf{£4,211}$

STEP 2: Annual cash flows when buying

Year 0: Purchase + Year-0 insurance + Year-0 maintenance
 $= -(\pounds 15,000 + \pounds 600 + \pounds 300) = \mathbf{-\pounds 15,900}$

Year 1: $-(\text{insurance Year1} + \text{maintenance Year1}) = -(\pounds 624 + \pounds 312) = \mathbf{-\pounds 936}$

Year 2: $-(\pounds 649 + \pounds 433) = \mathbf{-\pounds 1,082}$

Year 3: $-(\pounds 675 + \pounds 562) = \mathbf{-\pounds 1,237}$

Year 4: Resale receipt + $(-\text{insurance Year4} - \text{maintenance Year4}) = \pounds 3,000 - (702 + 702) = \pounds 3,000 - 1,404 = \mathbf{+\pounds 1,596}$

So the buying cash flow vector (Years 0→4) is:
 $\mathbf{[-15,900, -936, -1,082, -1,237, +1,596]}$

STEP 3: Discount each cash flow to present value at **9%**

- Discount factor for year $n = 1 / (1 + 0.09)^n$

Year	Cash flow (£)	Discount factor (9%)	Present value (£)
0	−15,900	1.0000	−15,900
1	−936	0.9174	−859
2	−1,082	0.8417	−911
3	−1,237	0.7722	−955
4	+1,596	0.7084	+1,131

Sum of PVs = **−£17,494** (this is the total present cost of buying).

STEP 4: Rent option: cash flows and PVs

Year	Rent cost (£)	Discount factor	PV (£)
0	−3,600	1.0000	−3,600
1	−3,744	0.9174	−3,435
2	−3,894	0.8417	−3,278
3	−4,050	0.7722	−3,127
4	−4,211	0.7084	−2,983

Annual Rent cash flows (Years 0→4) are negative amounts:

Rent cash flows: [−3,600 , −3,744 , −3,894 , −4,050 , −4,211]

Discount and compute PVs:

Sum of PVs = **−£16,423** (this is the total present cost of Renting).

STEP 5: Compare NPVs (present costs)

Buying: **–£17,494**

Renting: **–£16,423**

Because Renting has the **less negative** present value (i.e., a smaller total present cost), **hiring the 3D printer is the cheaper option** in present-value terms for this scenario. The difference is:

$$£17,494 - £16,423 = \mathbf{£1,071}$$

So Renting saves **£1,071** (in present-value terms) compared with buying.